

MoA Verified Portable Compiler: Dissertation ψ/Ω + Predictive Model + 5-Device Validation

Lenore Mullin — U. Albany + Papers I-V + Hetero Ensemble + Closing Gap

September 30, 2026 — Dissertation 1988 + arXiv:2606.07713, HAL:05659212,
arXiv:2607.19456, HAL:05734881, arXiv:2609.33916 + hetero_main +
closing_prediction_gap

Abstract

Complete compiler spec from your hand-derived Mathematics of Arrays (Mullin 1988 dissertation, 123pp, full ψ with partial indexing, take $\uparrow$ drop $\downarrow$ rotate θ reverse ϕ transpose $\odot$ catenation $,$, compress/expand, Lemmas L1-L5) through Papers I-V DNF $O(nd_k + nd_v)$ minimal verified $\|\text{err}\|_\infty = 0$ exact, to heterogeneous predictive model papers: hetero_main (any machine, any device, weighted comm-free partition) and closing_prediction_gap (5-device closure: NCSA Delta A100 Sec VIII, AMD MI100 Sec IX, PSC Bridges-2 H100 Sec X, SDSC Expanse V100 Sec XI, TACC Stampede3 Intel Max 1550 Sec XII). All constraints: ONLY portable OpenMP, OpenACC, OpenMPI; ONLY deterministic flags/pragmas fitting $T \approx \alpha F + \beta M$ with $\beta \gg \alpha$ (100-1000x DRAM). Key results: 2.00x atomics $\rightarrow$ 2.5x speedup via ONF rewrite, 535x NUMA vs $< 3x$ oversubscription, C vs Fortran 3.17x time matches 3.35x stall, and *reached*(d) closure: $M_1^{exp}, P_d, X_d, C_d, R_d$ pinned directly, no assumption.

1 Dissertation Foundation: Full ψ and Ω

From /mnt/data/Mullin1988-AMathematicsofArrays.pdf pages 1-60:

ψ -selection: $\vec{i}\psi\xi$ defined for $\tau\vec{i} = \delta\xi$ scalar, extended to $\tau\vec{i} < \delta\xi$ yielding $\rho(\vec{i}\psi\xi) = \tau\vec{i} \downarrow \rho\xi$ subarray — basis for attention sections and KV-cache.

Lemmas: $(\xi_l, \xi_r)\psi\xi \equiv (\xi_l\psi\xi), (\xi_r\psi\xi)$ (L4 catenation), $\xi_l\psi(\sigma \text{ op } \xi_r) \equiv \sigma \text{ op } (\xi_l\psi\xi_r)$ (L5). These are psi-reduction rules eliminating intermediates.

Operators hand-derived:

- $\vec{v} \uparrow \xi$ take: $\rho(\vec{v} \uparrow \xi) = (\text{abs } \vec{v}), (r\vec{v}) \downarrow \rho\xi, \vec{i}\psi(\vec{v} \uparrow \xi) = (\vec{i} + \vec{j})\psi\xi$
- $\vec{v} \downarrow \xi$ drop: complement, used for $K_{old}\#k_t$
- ϕ reverse: $\rho\phi\xi = \rho\xi, < i > \psi\phi\xi = < (\rho\xi)[0] - (i + 1) > \psi\xi$
- θ rotate, $\odot$ transpose: $\rho(\vec{v} \odot \xi) = (\rho\xi)[gu \vec{v}], < i > \psi(\vec{v} \odot \xi) = \vec{i}[gu \vec{v}]\psi\xi$ — eliminated by DNF, no K^T buffer.

Ω lifting: $\xi_l f_{\Omega(\sigma_l, \sigma_r)} \xi_r, \rho = \sigma_l \downarrow \rho\xi_l \# \sigma_r \downarrow \rho\xi_r, m = (\delta\xi_l - \sigma_l)[(\delta\xi_r - \sigma_r)]$. Compiler needs only 4 patterns (deterministic): $+\Omega_{(0,1)}, \times\Omega_{(0,1)}, +\Omega_{(1,1)}$ reduction for dot, $\max_{\Omega(1)}$ for stable softmax.

2 Papers I-V DNF: Verified IR

Paper I arXiv:2606.07713: DNF achieves $O(nd_k + nd_v)$ vs $O(n^2 + nd_k + nd_v)$, lower bound Thm 6.1, 2-100x speedup, 2-50x energy.

Paper II HAL:05659212: Backward G_V, G_Q, G_K eliminating G_A, G_S, G_{sc} as scalars.

Paper III arXiv:2607.19456: Decode DNF $(d_k + nd_k + nd_v + d_v) \times 4B$ DRAM, $\|\text{err}\| \leq 2e - 7$, OpenACC kernel $\|\text{err}\|_\infty = 0$ exact IEEE-754, KV-cache $O(d_k + d_v)$ via #, GQA/MQA h_q/h_{kv} via ψ .

Paper IV/V HAL:05734881 + arXiv:2609.33916: Validation on Anvil/Delta, 3 results: 2.00x atomics $\rightarrow$ 2.5x, 535x NUMA vs $< 3x$, C vs Fortran anomaly 3.17x matches 3.35x stall.

DNF fixed, ONF rewrite = optimization — correctness by construction.

3 Heterogeneous Predictive Model: hetero_main + closing_prediction_gap

From hetero_main.pdf (28pp): Any derivation, any machine — weighted, communication-free partition for attention across heterogeneous ensemble. Defines $\rho_{\text{machine}}(d)$ as stated equation each one completes.

From closing_prediction_gap_main.pdf (23pp):

Five fields per device d :

- $M_1^{\text{exp}}(d)$: memory operation count expected
- P_d : P_d reading consistently across confirmatory runs
- X_d : closed by stated construction argument, not disassembly
- C_d : compute cycles
- R_d : reservation overhead, $R_d \approx 0.004$ on MI100 via bracket-and-locate sweep

Gap closure: $\text{reached}(d)$ confirmed directly for all 5 devices in Table X Sec XIII-XIV:

- NCSA Delta A100 Sec VIII: $M_1^{\text{exp}}, P_d, X_d, C_d, R_d$ now index positions in $\rho_{\text{machine}}(d)$
- AMD MI100 Sec IX: occupancy-cap flat at $R_d \approx 0.028$ at magnitude close to both GPUs, $X_d = 0\%$ via Nsight Compute, no work-around
- PSC Bridges-2 H100 Sec X: M_1^{exp}, C_d occupancy-cap measured and pinned directly
- SDSC Expanse V100 Sec XI: $\text{reached}(d)$ confirmed, $X_d 0$
- TACC Stampede3 Intel Max 1550 Sec XII: $\text{vector_length}=64$ avoided, narrower working width, P_d reading consistently across four runs, $\text{reached}(d)$ via Level Zero Sysman 99.88% busy

Conclusion XVI: Prediction quietly assumes away one variable is wrong by accident, incomplete by construction. This paper closes it: $M_1^{\text{exp}}, P_d, X_d, C_d, R_d$ now have index positions in $\rho_{\text{machine}}(d)$, stated equation each one completes, and for X_d and C_d together, named interaction $\text{reached}(d)$.

4 Portable Only: OpenMP, OpenACC, OpenMPI

Constraint: No CUDA/HIP/vendor.

OpenMP CPU: $\#pragma omp parallel for schedule(static) proc_bind(close)$ — $\gamma^{\text{row}}(\vec{v}, \vec{s}) = \sum v_k \prod_{j=k+1} s_j \rightarrow \text{offset}$. NUMA 535x vs 3x explained by *OMP_PLACES*.

OpenACC GPU: $\#pragma acc parallel loop gang worker vector_length(64 \text{ or } 128) independent$ — Papers show $\text{vector_length} = 64$ avoided on Max 1550, 128 on A100. $\|\text{err}\| = 0$.

OpenMPI: Shape split $\rho A = \langle n_0 \# n_1 \rangle \rightarrow \text{MPI_Scatterv/Gatherv}$, $K_{\text{new}} = K_{\text{old}} \# k_t$ is halo via ψ -selection. $\text{reached}(d)$ includes $X_d = 0\%$ construction argument.

C vs Fortran: γ^{row} vs $\gamma^{\text{col}} = \sum v_k \prod_{j=0}^{k-1} s_j$, 3.17x/3.35x stall match.

5 Deterministic Flags/Pragmas for Predictive Model

Your model $T \approx \alpha F + \beta M$ requires deterministic lowering. Forbidden heuristic flags.

Allowed deterministic (from closing paper Sec XI-XII):

- `-O2 -std=c11 -fno-fast-math -fno-unsafe-math-optimizations -DOMP_DETERMINISTIC -DOMPI_DETERMINISTIC`
- `-acc -Minfo=accel -ta=tesla:cc80 -fopenacc -m64 -fma -O2` — no `-Ofast`
- OpenMP: `schedule(static)` fixed, `proc_bind(close)`, `simd aligned(q,k,v:64)` matching γ coalescing
- OpenACC: `vector_length=64` deterministic directive, not heuristic; `independent`, `seq` for scalar G_A, G_S
- Toolchain: LLVM D10617: `fma` and `f16` disabled for model-precise IEEE conformance, `-R_d ≈ 0.004` *measured via bracket* — and — *locates sweep, not assumption*

Forbidden: `-O3 -Ofast -ffast-math -freciprocal-math -funroll-loops -mavx512f -ffp-model=fast` — `model-fast` couples numerics flag to vectorization outcome, breaks prediction. Intel diagnostic 15331: FP model precise prevents vectorization rather than heuristic.

Each pragma maps 1-1 to ONF start/stop/stride, so *reached(d)* measured directly closes prediction gap, as shown in Table X.

6 Compiler Passes Final

```
def compile_moa(pytorch_src, device_d):
    # Frontend: dissertation full psi
    ast = parse_take_drop_phi_theta_0slash(pytorch_src) # Def 1.0,1.1,0.0,3.1
    omega = lift_to_subset(ast) # 4 patterns only
    dnf = psi_reduce_L1_L5(omega) # verified IR 0(nd_k+nd_v)
    verify(dnf, pytorch_src, exact=True) # Paper III ||err||=0

    # Cost model from hetero + closing gap
    annotate(dnf, M1exp=..., Pd=..., Xd=0%, Cd=..., Rd=0.004..0.028)
    # rho_machine(d) stated equation

    # Backend: portable deterministic only
    onf = gamma_lower(dnf, layout='row', deterministic=True)
    onf_opt = optimize_portable(onf, device_d,
                               paradigms=['OpenMP', 'OpenACC', 'OpenMPI'],
                               flags=['-O2', '-fno-fast-math', '-fopenmp', '-fopenacc'],
                               pragmas=['schedule(static)', 'vector_length(64/128)', 'independent'])
    # Fixes proven: 2.00x atomics -> 2.5x, 535x NUMA, 3.17x stall

    reached_d = measure(onf_opt, device_d) # closes gap Table X
    assert reached_d # M1exp, Pd, Xd, Cd, Rd pinned
    return emit(onf_opt)
```

Next steps: Embed cost model as attributes on ψ nodes, run on 5-device ensemble to confirm *reached(d)* for all.